The Parton Route to Quantum Spin Liquids

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Outline

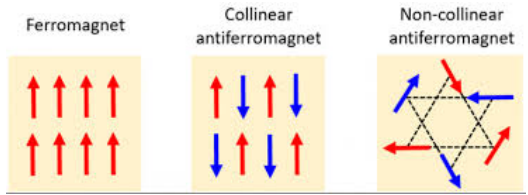
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Section 1. From Ordered Magnets to Spin Liquids

Magnetic order as a starting point

Ordinary magnets choose a pattern

- Ferromagnets: spins align
- Antiferromagnets: spins alternate
- Non-collinear magnets: spirals, stripes, textures



Ordered spin configurations

Once this pattern is known, low-energy physics is described by small fluctuations.

These are spin waves / magnons.

For ordered magnets, we know what to expand around.

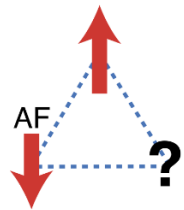
When spins refuse to order

Question

What if the ground state has no static spin pattern, even at $T = 0$?

Then the usual picture breaks down

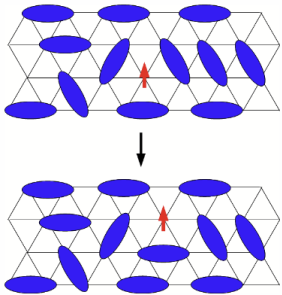
- No classical background
- No obvious order parameter
- No simple spin-wave expansion



Geometric frustration in an antiferromagnetic triangle

We need a language beyond ordinary magnetic order.

Quantum spin liquids: disorder without triviality



RVB picture: quantum superposition of singlet coverings

A quantum spin liquid is not a simple paramagnet.

Even at $T = 0$, the spins do not freeze into a fixed pattern.

Instead, the ground state remains a strongly correlated quantum superposition:

$$|\Psi_{\mathrm{QSL}}\rangle = \sum_{C} A_C \prod_{(ij)\in C} |s\rangle_{ij}$$

no static order $\neq$ no structure

The order is hidden in correlations, entanglement, and fractionalized excitations.

Ingredients of a quantum spin liquid

What is absent?

- No magnetic long-range order
- No fixed classical spin pattern
- No simple magnon-only description

What is present?

- Strong quantum fluctuations
- Long-range entanglement
- Fractionalized excitations
- Emergent gauge structure

A QSL is not merely disordered; it is a different kind of quantum order.

Why Partons?

Instead of expanding around an ordered spin pattern, we rewrite the spin using auxiliary fermions.

These auxiliary fermions are called **slave fermions**; their fractionalized excitations are called **spinons**.

$$\mathbf{S}_i \longrightarrow \text{slave fermions / spinons}$$

This makes fractionalization visible, but introduces extra unphysical states.



Cartoonish illustration

$$\text{enlarged Hilbert space} \xrightarrow{\text{constraint}} \text{physical spin states}$$

A hard spin problem becomes a constrained fermionic mean-field problem.

Slave Fermions and the Constraint

Spin as a fermion bilinear

$$S_i^{\mu} = \frac{1}{2} f_{i\alpha}^{\dagger} \sigma_{\alpha\beta}^{\mu} f_{i\beta}, \qquad \mu = x, y, z$$

Here $f_{i\uparrow}, f_{i\downarrow}$ are auxiliary fermions, often called **slave fermions** or **spinons**.

fermion space: $|0\rangle,\ |\uparrow\rangle,\ |\downarrow\rangle,\ |\uparrow\downarrow\rangle$ spin-$\frac{1}{2}$ space: $|\uparrow\rangle,\ |\downarrow\rangle$

So the physical Hilbert space is selected by the single-occupancy constraint:

$$\boxed{n_{i\uparrow} + n_{i\downarrow} = 1}$$

More generally, in the SU(2) slave-fermion language one also writes

$$\boxed{f_{i\uparrow}^{\dagger} f_{i\downarrow}^{\dagger} = 0, \qquad f_{i\downarrow} f_{i\uparrow} = 0}$$

Heisenberg Model in Slave-Fermion Form

Start with the antiferromagnetic Heisenberg model:

$$H = J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \qquad J > 0.$$

Using the slave-fermion identity,

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2} \left(f_{i\alpha}^\dagger f_{j\alpha} \right) \left(f_{j\beta}^\dagger f_{i\beta} \right) + \frac{1}{4} n_i n_j.$$

Since the physical subspace has $n_i = 1$, the density term is only a constant.

$$H \simeq -\frac{J}{2} \sum_{\langle ij \rangle} \left(f_{i\alpha}^\dagger f_{j\alpha} \right) \left(f_{j\beta}^\dagger f_{i\beta} \right) + \sum_i \lambda_i (n_i - 1).$$

Here λ_i enforces the single-occupancy constraint. Later, its fluctuations become part of the emergent gauge field.

The spin Hamiltonian becomes an interacting spinon Hamiltonian.

Mean-Field Decoupling: Two Natural Channels

$$H \simeq -\frac{J}{2} \sum_{\langle ij \rangle} \left(f_{i\alpha}^{\dagger} f_{j\alpha} \right) \left(f_{j\beta}^{\dagger} f_{i\beta} \right).$$

Mean field replaces spinon bilinears by bond averages.

Hopping channel

$$\chi_{ij} = \sum_{\alpha} \left\langle f_{i\alpha}^{\dagger} f_{j\alpha} \right\rangle$$

Spinons move through the lattice. This gives a spinon band structure.

Pairing channel

$$\Delta_{ij} = \langle f_{i\uparrow} f_{j\downarrow} - f_{i\downarrow} f_{j\uparrow} \rangle$$

Spinons form singlet pairs. This connects naturally to RVB physics.

$$H_{\mathrm{MF}} = H_{\mathrm{spinon}}(\chi_{ij}, \Delta_{ij}) + \sum_{i} \lambda_i (n_i - 1)$$

The interacting spin problem becomes a quadratic spinon mean-field problem.

Gauge Structure: The Price of Partons

The spin operator is unchanged under local transformations of the partons:

$$f_{i\alpha} \to e^{i\theta_i} f_{i\alpha}.$$

So the mean-field variables must transform too:

$$\chi_{ij} \to e^{i(\theta_i - \theta_j)} \chi_{ij}, \qquad \Delta_{ij} \to e^{i(\theta_i + \theta_j)} \Delta_{ij}.$$

Where gauge fields appear

$\lambda_i \quad \longrightarrow \quad$ temporal gauge field

$\arg(\chi_{ij})$, $\arg(\Delta_{ij}) \quad \longrightarrow \quad$ spatial gauge fields

Physical meaning

- Gauge freedom comes from redundancy.
- Only gauge-invariant quantities are physical.
- Pairing can reduce $U(1)$ gauge structure to $\mathbb{Z}_2$.

The parton theory is not just fermions; it is fermions coupled to an emergent gauge field.

From Heisenberg Partons to Quantum Spin Liquids

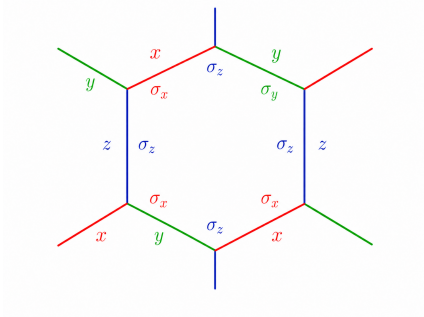
spin model $\longrightarrow$ slave fermions $\longrightarrow$ mean-field ansatz $\longrightarrow$ spin-liquid candidate

- The Heisenberg model gives the basic parton mean-field machinery.
- χ_{ij} describes spinon hopping.
- Δ_{ij} describes spinon pairing / RVB-like singlet formation.
- The constraint and gauge structure decide what is physically allowed.

This is the language we now use to understand more structured spin-liquid models.

Section 2. SU(2) Slave Fermion Solution of Kitaev Model

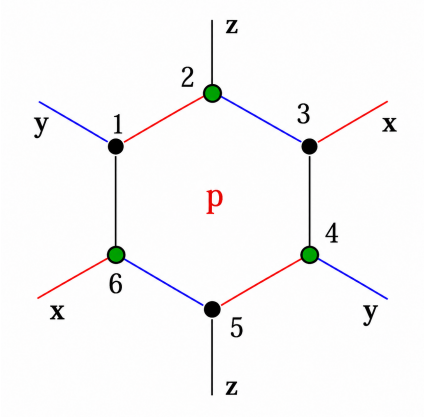
Kitaev Model on the Honeycomb Lattice



$$H = -J_x \sum_{\langle ij \rangle_x} S_i^x S_j^x - J_y \sum_{\langle ij \rangle_y} S_i^y S_j^y - J_z \sum_{\langle ij \rangle_z} S_i^z S_j^z$$

Red: x-links, Green: y-links, Blue: z-links

Conserved Quantities



For each plaquette p, define the plaquette operator:

$$\hat{W}_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z$$

$$[\hat{H}, \hat{W}_p] = 0 \qquad \forall\, p.$$

The set $\{W_p\}$ defines a particular flux sector.

Why is Kitaev Model a Quantum Spin Liquid?

The Kitaev model realizes a quantum spin liquid because bond-dependent spin interactions prevent conventional magnetic ordering and instead produce an exactly solvable, highly entangled ground state with fractionalized Majorana fermions and conserved Z_2 fluxes.

Hubbard–Stratonovich Decoupling

The Hubbard–Stratonovich transformation introduces auxiliary fields to rewrite an interaction term as a coupling to fluctuating fields:

$$\exp\left(-U n_\uparrow n_\downarrow\right) \longrightarrow \int \mathcal{D}\phi \, \exp\left[-\frac{\phi^2}{U} + \phi\left(n_\uparrow - n_\downarrow\right)\right].$$

For an x-type bond $\langle ij \rangle_x$, the decoupled Lagrangian depends on the hopping fields Φ_1, Φ_2 and pairing fields Θ_1, Θ_2:

$$\boxed{\mathcal{L}_x = \mathcal{L}_x\left(\Phi_1, \Phi_1^*, \Phi_2, \Phi_2^*, \Theta_1, \Theta_1^*, \Theta_2, \Theta_2^*, J_x, f_{i\sigma}, f_{i\sigma}^\dagger, f_{j\sigma}, f_{j\sigma}^\dagger\right)}$$

Mean Field Hamiltonian

To proceed, we assume that Φ_i and Θ_i acquire nonzero expectation values. We parametrize these expectation values by $t_{ij,\alpha}$ and $\Delta_{ij,\alpha}$, where $\alpha = \uparrow, \downarrow$.

$$H_{\mathrm{MF}} = \frac{1}{2}\sum_{\mathbf{q},\sigma} \psi^{\dagger}_{\mathbf{q}\sigma} \begin{bmatrix} 0 & t_{\sigma}(\mathbf{q}) & 0 & \Delta_{\sigma}(\mathbf{q}) \\ t^{*}_{\sigma}(\mathbf{q}) & 0 & -\Delta_{\sigma}(-\mathbf{q}) & 0 \\ 0 & -\Delta^{*}_{\sigma}(-\mathbf{q}) & 0 & -t^{*}_{\sigma}(-\mathbf{q}) \\ \Delta^{*}_{\sigma}(\mathbf{q}) & 0 & -t_{\sigma}(-\mathbf{q}) & 0 \end{bmatrix} \psi_{\mathbf{q}\sigma}.$$

Where

$$f_{\mathbf{q}\mu\sigma} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{R}\cdot\mathbf{q}} f_{\mathbf{R}\mu\sigma}.$$

Saddle Point Values of the Boson Fields

By imposing the self-consistency conditions on the mean-field expectation values, the hopping amplitudes $t_{ij,\alpha}$ and pairing amplitudes $\Delta_{ij,\alpha}$ can be written in terms of the Kitaev couplings J_x, J_y, and J_z. At the saddle point, the relevant momentum-dependent amplitudes are

$$t_{\uparrow}(\mathbf{q}) = -\frac{i}{16}\left(e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_1}J_x + e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_2}J_y\right),$$

$$\Delta_{\uparrow}(\mathbf{q}) = -\frac{i}{16}\left(e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_2}J_y - e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_1}J_x\right),$$

$$t_{\downarrow}(\mathbf{q}) = -\frac{i}{16}\left(e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_1}J_x + e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_2}J_y + 2J_z\right),$$

$$\Delta_{\downarrow}(\mathbf{q}) = \frac{i}{16}\left(e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_1}J_x + e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_2}J_y\right).$$

Spectrum of Mean Field Hamiltonian

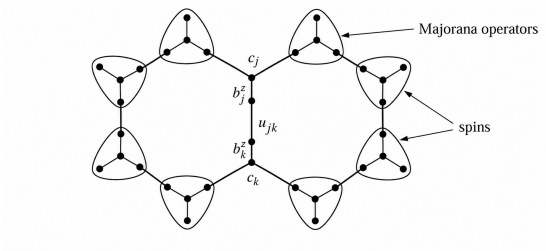
The band structure and eigenstates obtained from the mean-field Hamiltonian H_{MF} show that this formulation is consistent with the Majorana-fermion description of the Kitaev model. The mean-field spectrum contains three nondispersive, or flat, bands with energies

$$\epsilon_{\uparrow x} = \pm\frac{J_x}{8}, \qquad \epsilon_{\uparrow y} = \pm\frac{J_y}{8}, \qquad \epsilon_{\downarrow z} = \pm\frac{J_z}{8},$$

and one momentum-dependent dispersing band,

$$\epsilon_{\downarrow}(\mathbf{q}) = \pm\frac{1}{8}\left|J_x e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_1} + J_y e^{i\mathbf{q}\cdot\hat{\mathbf{l}}_2} + J_z\right|.$$

Connection with Majorana Fermions



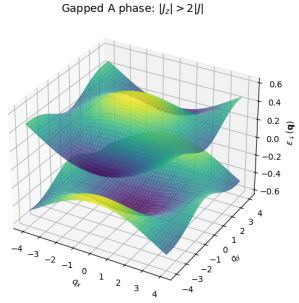
$$b^y_{\mathbf{q}\mu} = f^\dagger_{\mathbf{q}\mu\uparrow} + f_{-\mathbf{q}\mu\uparrow}, \qquad b^x_{\mathbf{q}\mu} = i\left(f^\dagger_{\mathbf{q}\mu\uparrow} - f_{-\mathbf{q}\mu\uparrow}\right),$$

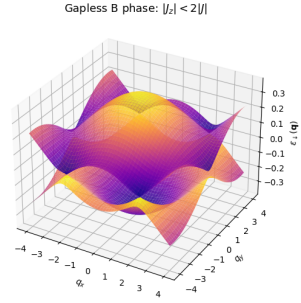
$$b^z_{\mathbf{q}\mu} = f^\dagger_{\mathbf{q}\mu\downarrow} + f_{-\mathbf{q}\mu\downarrow}, \qquad c_{\mathbf{q}\mu} = i\left(f^\dagger_{\mathbf{q}\mu\downarrow} - f_{-\mathbf{q}\mu\downarrow}\right).$$

Slave Fermion Band Structure

To understand the physics of this model, it is useful to focus on the band structure of the down-spin fermions. It suffices to consider the case $J_x = J_y = J$:

$$\epsilon_{\downarrow}(\mathbf{q}) = \pm\frac{J}{8}\left\{\left(\frac{J_z}{J} + 2\cos\frac{q_x}{2}\cos\frac{\sqrt{3}q_y}{2}\right)^2 + 4\left(\cos\frac{q_x}{2}\sin\frac{\sqrt{3}q_y}{2}\right)^2\right\}^{1/2}.$$





Phases of the Solutions

Table: Comparison of the topological and mean-field character of the gapped A phase and gapless B phase.

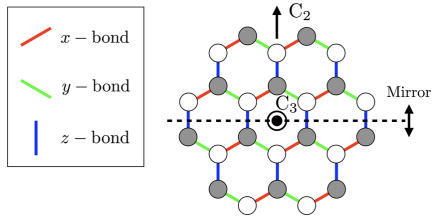
Feature	**Gapped A Phase**	**Gapless B Phase**
Topological nature	Abelian topological phase.	Gapless spin-liquid phase; becomes non-Abelian when time-reversal symmetry is broken.
Mean-field interpretation	Fully gapped p-wave superconducting state of slave fermions.	Gapless p-wave superconducting state of slave fermions.
Effect of time-reversal symmetry breaking	Remains gapped and Abelian in the usual A-phase regime.	The Dirac fermion acquires a mass, producing a gapped non-Abelian phase.

Kitaev–Heisenberg Hamiltonian

Minimal model on the honeycomb lattice

$$H_{ij}^{(\gamma)} = -J_1 S_i^\gamma S_j^\gamma + J_2\, \mathbf{S}_i \cdot \mathbf{S}_j$$

- $S_i^\gamma S_j^\gamma$: bond-dependent Kitaev exchange
- $\mathbf{S}_i \cdot \mathbf{S}_j$: isotropic Heisenberg exchange
- $\gamma = x, y, z$: three inequivalent bond directions



Bond-dependent exchange and schematic phases.

$$J_1 = 2\alpha, \qquad J_2 = 1 - \alpha, \qquad 0 \le \alpha \le 1$$

Interpolating Hamiltonian

$$H = \sum_{\langle ij \rangle_\gamma} \left[(1-\alpha)\, \mathbf{S}_i \cdot \mathbf{S}_j - 2\alpha\, S_i^\gamma S_j^\gamma \right]$$

Parton Mean Field Theory of Kitaev Heisenberg Model

$$E^{\mu}_{i,p} = \frac{1}{2} f^{\dagger}_{i+p,\alpha} \sigma^{\mu}_{\alpha\beta} f_{i\beta},$$

$$D^{\mu}_{i,p} = \frac{1}{2} f_{i+p,\alpha} \left(i\sigma^{y}\sigma^{\mu} \right)_{\alpha\beta} f_{i\beta}.$$

$$S^{\alpha}_{i} S^{\alpha}_{i+p} = \frac{1}{2} \sum_{\beta=x,y,z} (1 - \delta_{\alpha\beta}) \left[\left(E^{\beta}_{i,p} \right)^{\dagger} E^{\beta}_{i,p} + \left(D^{\beta}_{i,p} \right)^{\dagger} D^{\beta}_{i,p} \right] - \frac{n_i}{4}.$$

Away from the Kitaev limit, the Heisenberg term competes with the spin-liquid amplitudes. In mean-field language, the competition is between E, D, and the magnetic order parameter m. When $m = 0$, the ansatz has a $\mathbb{Z}_2$ invariant gauge structure and describes the Kitaev-connected spin liquid. When $m \neq 0$, the state develops ferromagnetic order in the rotated basis, which is the stripy antiferromagnet in the original basis.

Section 3. Numerics on the Kitaev–Heisenberg Model

KitaevâHeisenberg Model: Numerical Task

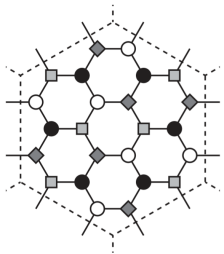
Numerical task

Track the finite-cluster ground state as α is tuned.

Starting point

$$H = \sum_{\langle ij \rangle_\gamma} \left[(1-\alpha)\, \mathbf{S}_i \cdot \mathbf{S}_j - 2\alpha\, S_i^\gamma S_j^\gamma \right], \qquad 0 \le \alpha \le 1.$$

- $\alpha = 0$: antiferromagnetic Heisenberg limit
- $\alpha = 1$: bond-dependent Kitaev limit
- intermediate α: competition between magnetic order and Kitaev physics



Four-sublattice/Klein pattern used for the rotated-basis diagnostics.

$$\widetilde{S}_i^a = \eta_i^a S_i^a, \qquad a = x, y, z$$

The four-sublattice signs $\eta_i^a = \pm 1$ define the rotated spin basis. They flip selected spin components so that stripy order appears as hidden ferromagnetic order.

Rotated basis: making stripy order visible

Four-sublattice sign structure

sublattice	η^x	η^y	η^z
I	$+$	$+$	$+$
X	$+$	$-$	$-$
Y	$-$	$+$	$-$
Z	$-$	$-$	$+$

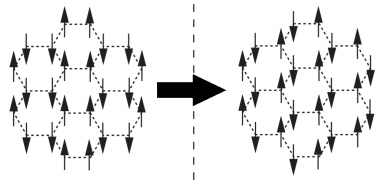
The transformation flips selected spin components:

$$\widetilde{S}_i^a = \eta_i^a S_i^a.$$

At the special point,

$$\widetilde{H}_{\alpha=1/2} = -\frac{1}{2}\sum_{\langle ij\rangle} \widetilde{\mathbf{S}}_i \cdot \widetilde{\mathbf{S}}_j.$$

At $\alpha = \frac{1}{2}$, the rotated Hamiltonian is ferromagnetic.

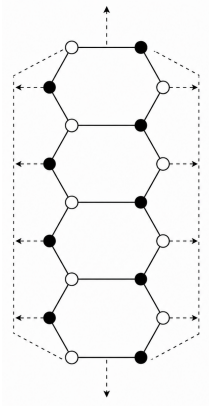


Stripy order in the original basis becomes ferromagnetic in the rotated basis.

Key point

The stripy AF phase is a hidden ferromagnet after the four-sublattice rotation.

Finite-cluster exact diagonalization



Periodic 16-site honeycomb cluster

Cluster used for the calculation

$$N = 2L_xL_y, \qquad L_x = 2, \quad L_y = 4, \qquad N = 16$$

$$\dim \mathcal{H} = 2^N = 2^{16} = 65536$$

Nearest-neighbour bond count

$$N_b = 24, \qquad N_b^x = N_b^y = N_b^z = 8$$

The equal number of x, y, and z bonds keeps the finite cluster balanced with respect to the three Kitaev bond directions.

Ground-state problem

$$\boxed{\mathrm{H}(\alpha)\psi_0(\alpha) = E_0(\alpha)\psi_0(\alpha)}$$

Why matrix-free Lanczos?

Dense matrix route

$$D^2 = (65536)^2 \simeq 4.29 \times 10^9$$

dense-matrix entries

Problem

Storing and diagonalizing the full many-body matrix is impractical.

Matrix-free route

$$v \longmapsto Hv$$

basis state $\rightarrow$ bond action $\rightarrow$ new vector

Only the local bond action is applied repeatedly; the full Hamiltonian matrix is never stored.

Lanczos step

$$\mathcal{K}_m = \mathrm{span}\{v_0, Hv_0, \ldots, H^{m-1}v_0\}$$

Extract $E_0(\alpha)$ and $\psi_0(\alpha)$ from the Krylov subspace.

Dense matrix impossible $\rightarrow$ use Hv $\rightarrow$ Lanczos $\rightarrow$ ground state.

Diagnostics measured from the ground state

Total spin

$$\frac{\langle S_{\mathrm{tot}}^2 \rangle}{S_{\mathrm{FM}}(S_{\mathrm{FM}}+1)}$$

Measures ordinary magnetic character in the original spin basis.

Nearest-neighbour correlations

$$C_{\mathrm{NN}}, \qquad \widetilde{C}_{\mathrm{NN}}, \qquad C_K$$

Compares original antiferromagnetic correlations, rotated-basis correlations, and Kitaev bond-matching correlations.

Rotated total spin

$$\frac{\langle \widetilde{S}_{\mathrm{tot}}^2 \rangle}{S_{\mathrm{FM}}(S_{\mathrm{FM}}+1)}$$

Detects hidden ferromagnetic order, useful for identifying the stripy phase.

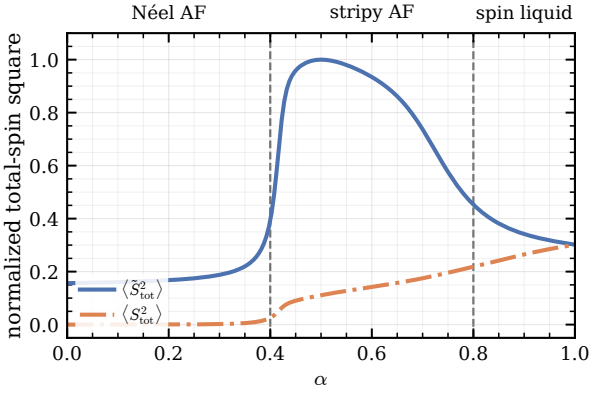
Energy and field response

$$-\partial_\alpha^2 E_0, \qquad m, \qquad \chi^{-1}$$

Locates rapid ground-state changes and tests the stability of hidden order under a weak rotated field.

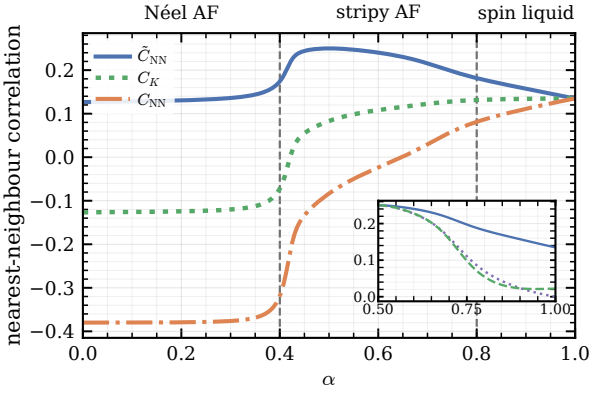
Together, these diagnostics separate the Néel region, the stripy hidden-ferromagnetic region, and the Kitaev-like regime.

Total-spin diagnostic



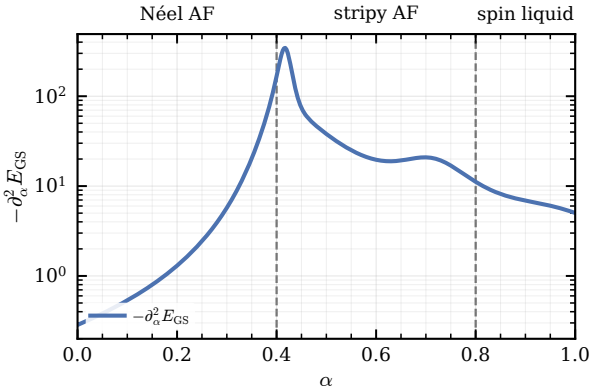
The rotated total spin nearly saturates in the intermediate region, identifying the stripy phase as a hidden ferromagnet.

Nearest-neighbour correlations



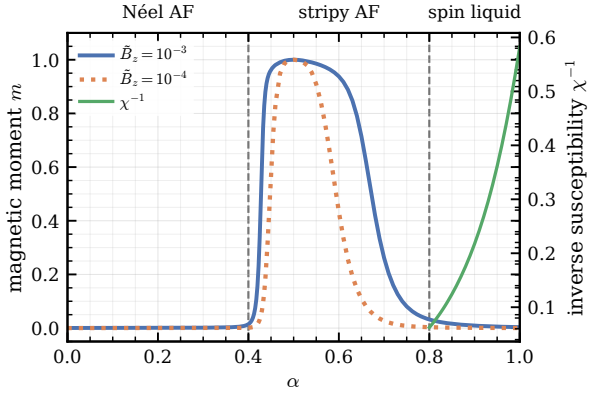
Original-basis correlations are antiferromagnetic near $\alpha = 0$, while rotated-basis correlations become positive in the stripy region.

Ground-state energy curvature



The sharp peak near $\alpha \simeq 0.4$ marks the strongest ground-state reorganization; the broader feature near the Kitaev side is smoother.

Weak-field response



The rotated-field response is large in the stripy region and collapses toward the Kitaev limit, where χ^{-1} rises.

Takeaway

Néel AF $\longrightarrow$ stripy AF / hidden FM $\longrightarrow$ Kitaev-like

Finite-cluster exact diagonalization reproduces the qualitative phase sequence of the Kitaev–Heisenberg model.

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